

ARUN NATARAJAN

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ACADEMIC PROFILE

Centennial High School (CHS) — Frisco, TX

Class Rank: 28/502 | Unweighted GPA: 3.967/4.0 | Weighted GPA: 5.367

SAT: 1530 (Math: 800, Evidence-Based Reading & Writing: 730)

Dual Credit Coursework (Collin College): Calculus III (Multivariable Calculus), Differential Equations

Senior Year AP Coursework: AP Physics C (Mechanics and Electricity & Magnetism), AP English Literature, AP Macroeconomics, AP U.S. Government, AP Microeconomics, AP Psychology

Completed AP Exams (Scores): Chemistry (5), Calculus BC (5), Physics 1 (5), Statistics (5), Computer Science A (5), Computer Science Principles (5), English Language & Composition (5), Biology (4), U.S. History (4), World History (4), Human Geography (4), Seminar (3)

HONORS AND AWARDS

- **3rd Place Nationally**, Health Administration Concepts — Business Professionals of America (11th)
- **3rd Place Nationally**, SQL Database Fundamentals — Business Professionals of America (11th)
- **International Career Development Conference Qualifier (ICDC)**, Virtual Business Challenge — DECA (11th)
- **6th & 11th Place Statewide**, UIL 5A Best of Texas Chemistry (11th)
- **AP Scholar with Distinction**, College Board (10th and 11th)
- **9th Place Nationally**, SQL Database Fundamentals — Business Professionals of America (10th)

ENGINEERING PROJECTS AND RESEARCH

UC San Diego Biomolecular Research Program

Student Researcher | 12th

- Conducted 100 hours of independent literature research on damage-associated molecular patterns (DAMPs) and mechanisms of cartilage degradation.
- Analyzed published findings on cartilage explants, tracing how fibronectin fragments increase matrix metalloproteinase production in cartilage cells.
- Synthesized the findings in an 8-page research paper and presented and defended the analysis before university faculty and program peers.

Arduino Bioreactor Temperature Control Simulation

Independent Project | 11th

- Developed an Arduino simulation in Tinkercad to model closed-loop temperature control for mammalian cell culture in a bioreactor.
- Programmed feedback logic with simulated thermistors and relay interlocks, maintaining a simulated temperature of $37.0^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$.

Arduino Distillation Column Simulator

Independent Project | 11th

- Developed a Tinkercad process simulation to model temperature profiles across the trays of a multicomponent distillation column.
- Implemented calculations using simulated reboiler sensor readings to visualize vapor-liquid equilibrium (VLE) across separation stages.

UT Dallas Computational Biology Workshop

Participant / 10th

- Modeled viral epitopes with structural bioinformatics tools and estimated antibody-antigen binding affinities.
- Simulated antibody-antigen binding through molecular docking to investigate potential cross-neutralization across viral variants.

Interactive Chemistry Lab – 3D Molecular Modeling Platform

Lead Developer / 10th

- Developed an interactive WebGL platform for exploring molecular geometry, bond configurations, and three-dimensional rotation.
- Implemented real-time controls and molecular renderers showing the 104.5° bond angle in water (H₂O) and the pyranose ring structure of glucose.

LEADERSHIP AND SERVICE

BioMolecular Insights

Founder & Content Creator / 11th–12th

- Created 12+ educational videos, researching, scripting, and animating explanations of biofluid dynamics, circulatory mechanics, enzyme kinetics, and pharmacokinetics.
- Reached 3,000+ views with biomolecular engineering content for high school and undergraduate audiences.

DoGoodHub.org

Founder & Lead Developer / 11th–12th

- Developed and launched a full-stack website connecting high school students with local volunteer opportunities.
- Designed a searchable relational database indexing 50+ food banks, free healthcare clinics, emergency shelters, and legal aid centers.

Tessera Applications LLC

Student Intern / 11th–12th

- Built responsive mobile UI screens and connected frontend components to backend APIs for automated document and text processing.
- Conducted functional testing, wrote bug fixes, and assisted in shipping app updates to improve performance and stability.

Carrollton Regional Medical Center (CRMC)

Clinical Volunteer / 11th

- Selected as one of 15 regional students for a limited-capacity, 100-hour hospital immersion program.
- Observed ICU rounds, arterial blood gas analysis, and mechanical ventilation monitoring while shadowing respiratory therapy teams.

Distributive Education Clubs of America (DECA)

Varsity Competitor & International Finalist / 10th–12th

- Selected from 400+ chapter members to represent the school delegation across state and international competitive conferences.
- Qualified for the ICDC Virtual Business Challenge by analyzing player return on investment, stadium revenue, and overall franchise profitability.

Business Professionals of America (BPA)

Mentorship Officer & National Competitor | 9th–12th

- Led weekly technical study sessions for 20+ club members covering database concepts, SQL fundamentals, and presentation skills.
- Developed practice problem sets and diagnostic reviews, helping multiple peers qualify for state and national competitions.

MedSafety

Founder & Lead Developer | 10th

- Identified OTC dosage and language barriers at two community pharmacies in Chennai and created a bilingual safety guide used by pharmacists serving 100+ patrons daily.
- Developed and launched a free English-Tamil web application covering OTC uses, contraindications, and pediatric dosage information.

UT Southwestern Medical Center (UTSW) — HPREP

Selected Scholar & Participant | 10th

- Selected through a competitive regional application process for an intensive healthcare symposium hosted by UTSW clinical faculty and medical students.
- Participated in faculty-led clinical diagnostics, bioethics case discussions, and translational medicine seminars exploring healthcare delivery and biotechnology.

i9 Sports

Youth Referee & Coach | 9th–10th

- Coached and mentored 50+ youth athletes per season in soccer, basketball, baseball, and flag football.
- Officiated games as lead referee, applying rules, maintaining safe play, and encouraging sportsmanship.

TECHNICAL SKILLS AND CERTIFICATIONS

Certifications: The Jackson Laboratory (CRISPR/Cas9 Gene Editing, Introduction to Genetic Engineering); YouScience (Biotechnology, Computer Science Principles, Computer Science II Java); Certiport IT Specialist (Databases)

Applied Concepts: Closed-Loop Control, Vapor-Liquid Equilibrium, Molecular Docking, Pharmacokinetics

Programming & Tools: Python, Java, SQL, Arduino IDE, Tinkercad Circuits, Git